THE SHEAF ASSOCIATED TO A QUASI-COHERENT MODULE IS QUASI-COHERENT ON THE BIG AND ON THE SMALL SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement quotes the definition of quasi-coherence on a site by reference to Modules on Sites; the text of that definition is not reproduced and the dependency is recorded as a standard prerequisite.

Editorial note. The counted spans carry 3 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Quasi-coherent sheaves of modules on a ringed space and their local presentation. Modules, Definition modules-definition-quasi-coherent, invoked at proof lines 2639-2641. Supplies the Zariski cover of S and the index sets over which F is presented by direct sums of the structure sheaf.
- The definition of a quasi-coherent module on a site. Modules on Sites, Definition sites-modules-definition-site-local, quoted in the statement at lines 2622-2628. Fixes both conclusions, on the big site and on the small Zariski and etale sites.
- Quasi-coherence on a site is local for a covering. Modules on Sites, Lemma sites-modules-lemma-local-final-object, invoked at proof lines 2655-2656 and 2669-2670. Lets the presentation over each member of the Zariski cover be used to conclude quasi-coherence of F$\widehat{\ }$a.

Source reference. Original definition 01BE: sources/stacks-chow/tags.txt, source lines 1723–1723. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 03DL: sources/stacks-chow/tags.txt, source lines 4424–4424. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 03DN: sources/stacks-chow/tags.txt, source lines 4426–4426. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_S$-module. Let $\tau \in \{Zariski, étale, smooth, syntomic, fppf\}$.*

1. *The sheaf $\mathcal{F}^a$ is a quasi-coherent $\mathcal{O}$-module on $(Sch/S)_\tau$, as defined in Modules on Sites, Definition 03DL.*
2. *If $\tau = Zariski$ or $\tau = étale$, then the sheaf $\mathcal{F}^a$ is a quasi-coherent $\mathcal{O}$-module on S_{Zar} or $S_{étale}$ as defined in Modules on Sites, Definition 03DL.*

Proof. Let $\{S_i \to S\}$ be a Zariski covering such that we have exact sequences

$$\bigoplus\nolimits_{k\in K_i} \mathcal{O}_{S_i} \longrightarrow \bigoplus\nolimits_{j\in J_i} \mathcal{O}_{S_i} \longrightarrow \mathcal{F}|_{S_i} \longrightarrow 0$$

for some index sets K_i and J_i. This is possible by the definition of a quasi-coherent sheaf on a ringed space (See Modules, Definition 01BE).

Proof of (1). Let $\tau \in \{Zariski, fppf, étale, smooth, syntomic\}$. It is clear that $\mathcal{F}^a|_{(Sch/S_i)_\tau}$ also sits in an exact sequence

$$\bigoplus\nolimits_{k\in K_i} \mathcal{O}|_{(Sch/S_i)_\tau} \longrightarrow \bigoplus\nolimits_{j\in J_i} \mathcal{O}|_{(Sch/S_i)_\tau} \longrightarrow \mathcal{F}^a|_{(Sch/S_i)_\tau} \longrightarrow 0$$

Hence $\mathcal{F}^a$ is quasi-coherent by Modules on Sites, Lemma 03DN.

Proof of (2). Let $\tau = étale$. It is clear that $\mathcal{F}^a|_{(S_i)_{étale}}$ also sits in an exact sequence

$$\bigoplus\nolimits_{k\in K_i} \mathcal{O}|_{(S_i)_{étale}} \longrightarrow \bigoplus\nolimits_{j\in J_i} \mathcal{O}|_{(S_i)_{étale}} \longrightarrow \mathcal{F}^a|_{(S_i)_{étale}} \longrightarrow 0$$

Hence $\mathcal{F}^a$ is quasi-coherent by Modules on Sites, Lemma 03DN. The case $\tau = Zariski$ is similar (actually, it is really tautological since the corresponding ringed topoi agree). □